

KEVIN LEVY

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PROFILE

Quantitative Scientist with a PhD in Astrophysics, specializing in statistical inference, machine learning, and scientific computing. Experienced building end-to-end data pipelines and HPC workflows, from raw data acquisition to statistical modeling and uncertainty quantification. Interested in research applying machine learning to environmental science, particularly using remote sensing datasets.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib), SQL, Shell Scripting
- **Data Analysis:** Data Cleaning/Validation/Visualization, Data Pipeline Development, Fourier & Time Series Analysis
- **Statistics & Machine Learning:** Frequentist & Bayesian Inference, MCMC Sampling, Supervised & Unsupervised Learning, Neural Networks
- **Cloud Platforms:** Google Cloud Platform (GCP; Vertex AI, BigQuery, Looker, Cloud Storage)
- **Infrastructure & Computing:** Linux, High-Performance Computing (HPC), Docker, Git/GitHub

PROFESSIONAL EXPERIENCE

Doctoral Researcher, Astrophysics — University of Melbourne

Jun 2022 – Jun 2026

- **Scientific Impact:** Led the development of two end-to-end Python pipelines:
 1. Reconstructed and statistically analyzed the first projected matter distribution of the Universe from South Pole Telescope Summer survey data, confirming the data quality is sufficient for precise cosmological measurements;
 2. Created unbiased estimates of galaxy cluster masses from millimeter-wave data. The pipeline was later used to obtain the most constraining millimeter-wave cluster mass measurements to date (~2x precision improvement).
- **Validation:** Ran a suite of systematic tests to control for bias and ensure robust, reproducible results.
- **Collaboration & Communication:** Collaborated within an international research team on shared, version-controlled codebases and presented findings at international conferences.
- **Publications:** [Levy et al. 2026](#), [Levy et al. 2023](#)

BI Developer — Delta IT S.A.

Dec 2021 – May 2022

- **Data Engineering & Visualization:** Used SQL to build and maintain database views integrated with the WIGES ERP platform, developing dashboards and charts to track financial, buying, and selling KPIs across the client base.
- **Stakeholder Management:** Worked directly with clients on-site and via calls/Zoom to gather requirements, customizing dashboards to match each client's specific needs and preferences.

INDEPENDENT PROJECTS

Flood Detection from Satellite Imagery

- Built an end-to-end remote sensing pipeline for flood detection by training a PyTorch U-Net from Sentinel-1 SAR imagery, achieving 79% mean IoU (segmentation accuracy).
- Deployed the model as a live, publicly callable API on Google Cloud Run, using a custom containerized training job on Vertex AI for scalable training.
- Built a companion analysis integrating ERA5 reanalysis data to examine climate correlations with flood events, connecting remote sensing outputs to atmospheric reanalysis products.

Bayesian Model Selection for Galaxy Distance Distributions

- Built an automated Python pipeline fitting Gaussian Mixture Models to noisy galaxy distance distribution data via MCMC, using BIC for data-driven model-order selection.
- Designed an iterative reweighting scheme to jointly calibrate a bias correction across multiple data subsets.

EDUCATION

- Ph.D. in Astrophysics — University of Melbourne Jun 2022 – Jun 2026
- M.Sc. in Astrophysics — University of Bonn Oct 2019 – Oct 2021
- B.Sc. in Physics — University of Bonn Oct 2015 – Oct 2019

CERTIFICATIONS & AWARDS

- Google Cloud Computing Foundations Certificate — Google (2026)
- Data Scientist in Python Path — Dataquest.io (2021)
- N.D. Goldsworthy Scholarship (2022) — Awarded on the basis of academic excellence

LANGUAGES

English (fluent) · French (fluent) · German (fluent) · Luxembourgish (fluent)